

JAKUB KIERZNOWSKI

Junior Software Engineer | Software Developer Intern at IBM
CS & Intelligent Systems Student at AGH

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github.com/qualv13 | linkedin.com/in/jakub-kierznowski | qualv13.github.io | instaldesk.pl

Backend engineer working in Java and Python, drawn to graph data and to code that ships with tests. At IBM I optimise graph and tree traversal in Data Lineage over enterprise metadata. After hours I build and support InstalDesk, a live B2B SaaS with a native integration to the Polish KSeF e-invoicing system. Final-year Computer Science and Intelligent Systems at AGH, GPA 4.74/5.00, graduating January 2027.

Java 21 · Spring Boot · Spring Security · PostgreSQL · Neo4j · Python · FastAPI · Docker · Testcontainers · Jenkins

EXPERIENCE

Software Developer Intern

IBM Software Lab · Kraków · hybrid

Part-time

Feb 2026 – Present

- Java and Spring on Data Lineage: graph and tree traversal over large enterprise metadata graphs. Scale and internals are covered by IBM confidentiality.
- Built the automated pytest suite for that pipeline — parametrised cases, session-scoped fixtures, async execution. Cut the run from 20 minutes to 4.
- Own the Jenkins job scheduling for the pipeline.
- Daily Scrum in an English-speaking cross-functional team.

IT Team Lead, Web & Workflow Automation

Student Council · WRSS WEAlilB, AGH

Mar 2025 – Present

Council member since Mar 2024, leading its IT team since Mar 2025. I maintain the council website (Vue, TailwindCSS) and the VPS it runs on, and built internal automations on self-hosted n8n against Slack and Google APIs over OAuth 2.0. Event coordinator for the VOLT 2025 tech conference, eight partner companies including Aptiv, PwC Polska and Cytiva, and part of the organising team for VOLT 2026.

Team Mentor and Community Leader

Zwolnieni z Teorii · remote

Sep 2022 – May 2025

Mentored 35+ student teams per edition, from planning to the final presentation of their social-impact projects. Later ran a participant community (ZWzT × PwC certificate) and supported olympiad entrants.

Game Developer

AGH Code Industry · Dec 2023 – Jun 2024

Gameplay mechanics in Unity and C# for “Student Trainer”, a game built by a student team under a mentor.

PROJECTS

Code at [<repo>](https://github.com/qualv13)

InstalDesk, B2B SaaS for Polish installation contractors

instaldesk.pl · live since Apr 2026, v1.9.0, pilot users · Python, FastAPI, PostgreSQL, SQLite, Stripe, KSeF API

- Native KSeF integration: API v2 against the FA(3) schema, KSeF-token auth on web, XAdES-BES certificate auth on desktop — filed straight to the Ministry of Finance, without a third-party integrator.
- A web PWA and a Windows desktop client share one account. PostgreSQL in the cloud, SQLite locally so the desktop client keeps working offline.
- Per-tenant isolation, KSeF tokens encrypted at rest. Stripe billing over card, BLIK and Przelewy24, a 14-day trial without a card, renewal flows, transactional email, PDF generation, XLSX and CSV import and export.
- Built and supported solo, including user contact.

IoT fleet backend

IoTServerApp · Java 17, Spring Boot 3.3, PostgreSQL, RabbitMQ/MQTT, Protocol Buffers, Docker

Layered Spring backend for managing smart-lamp fleets, with 15 JUnit 5 test classes across controllers, services, repositories and security, and integration tests on Testcontainers. JWT auth with refresh tokens, MQTT command and telemetry topics over RabbitMQ, a Protocol Buffers device protocol, OTA firmware delivery from S3, scheduled device jobs. Full Docker Compose stack.

DrugPath, biomedical knowledge-graph agent

neo4j-agent · Neo4j Aura Agent Hackathon 2026, top 6 · Neo4j Aura, Cypher, Python, LLM agents

An agent that walks the Hetionet biomedical graph (47,031 nodes, 293k relationships) along drug → gene → pathway → disease paths to explain how a drug works and where it might be repurposed. Educational project on public research data. Demo at qualv13.github.io/neo4j-agent.

AI-native hardware rental and inventory hub

hardware-hub · Booksy technical assessment · Python, FastAPI, Vue, Gemini, Docker

Internal tool for tracking equipment, availability and hand-offs: semantic search over the inventory, an auditor that flags inconsistent records, and Gemini tool-calling for repairs. Every AI call has a deterministic fallback, so the product still works when the model does not.

BPM-first workout playlist generator

Fit2Beat · HackYeah 2025, top 8 of 65+ · Python, Flask, Docker

Builds playlists from a target BPM rather than from genre. Nobody else in the biohacking track approached it that way. Team of four, 24 hours, 3.9/5 from the jury.

Green-window EV charging optimiser

CarCharging · Java, Spring Boot, REST, Docker

Finds the cleanest 48-hour charging window from the UK Carbon Intensity API with a sliding window over half-hourly forecasts, and serves the forecast through a REST API.

Graph and deep neural networks

GGSN · Graph and Deep Neural Networks, AGH coursework · PyTorch, TensorFlow, scikit-learn

Graph neural networks (GCN, GAT, GraphSAGE, GIN) over the Yelp dataset, benchmarked against one another. Transfer learning with Grad-CAM explanations and DistilBERT sentiment classification. Network compression by pruning, quantisation and distillation across seven architectures, compared on accuracy against model size.

SKILLS

Tech

Core: Java 21, Spring Boot, Spring Security, Hibernate/JPA, Maven, JUnit 5, Testcontainers, PostgreSQL, SQL, REST APIs.

Working knowledge: Python (FastAPI, pytest), Docker and Compose, Linux/Bash, Git, Jenkins CI/CD, RabbitMQ, MQTT, Protocol Buffers, OAuth 2.0, Neo4j and Cypher, AWS S3, algorithms and data structures.

Also used: Vue with TailwindCSS, C# (Unity), C++, n8n, LLM APIs, self-hosted models (Ollama), TensorFlow, scikit-learn.

Soft

Agile and Kanban, daily Scrum in a cross-functional English-speaking team. Lead event coordinator for the VOLT 2025 conference, including partnership deals with tech companies. I ship my own product and handle its support and user contact myself.

Languages: Polish native, English C1 (working language at IBM), French A2.

EDUCATION

Computer Science and Intelligent Systems, BEng

AGH University of Krakow · Kraków
Oct 2023 – Jan 2027

GPA 4.74/5.00. Year Representative since the 1st semester, Student Council since the 2nd. Engineering thesis: applying deep learning to 2D-image to 3D-mesh conversion.

Extended Mathematics, Physics and Computer Science

V High School · Kraków
2019 – 2023

C++, competitive-programming problem solving.

CERTIFICATIONS

Neo4j Certified Professional and Building Agents in Neo4j Aura, both 2026.

Amazon Junior Software Developer (Coursera specialisation): Data Structures and Algorithms, Programming with Java, Application Development.

INTERESTS

Snowboarding, sailing, a self-hosted homelab with Docker and quantised local models on a Raspberry Pi and a VPS, game dev, board games.